Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years shipping **user-facing applications** across **React**, **Java**, and **Android/iOS** clients. Strong in **low-latency concurrency**, networking, and OS fundamentals; experienced iterating with end users on performant, reliable control interfaces.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Computer Networks, Analysis of Algorithms, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Data Structures, Object-Oriented Programming, Computer Networks

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected full-stack platform services for 500+ users, cutting dashboard latency from **1.2s to 350ms** through **concurrent async workers**, query tuning, and memory-efficient data access patterns.
- Shipped production features with fast iteration cycles, code/design reviews, and distributed-system debugging to maintain **99%+ reliability** across cross-platform user workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **React + Flask** user-facing grading software for 150+ instructors, iterating on UX feedback while handling **10K+ daily requests** with **multithreaded** batch processing and stable API performance.
- Reduced instructor feedback turnaround by **60%** through rapid feature delivery, technical presentations, and cross-functional scoping with product and design partners.

USC AD Labs **Los Angeles, CA**
Software Engineering Assistant *January 2025 – May 2025*

- Developed high-throughput processing services across 100K+ records with reproducible validation, design reviews, and clear communication in architecture discussions.
- Contributed iterative improvements and documentation to strengthen engineering quality in multidisciplinary research delivery.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Java/Python microservices** with low-latency networking and integration flows on **AWS**, improving response time by **30%** through maintainable, performant service design.
- Led code reviews and **20+ automated test suites** in Jenkins CI/CD, increasing release confidence by **68%** across Windows/Linux distributed environments.

LST Capitals **India**
Software Engineer Intern *November 2022 – January 2023*

- Delivered **React Native Android/iOS** and web clients with **Flask APIs**, achieving **sub-200ms** booking flows via Firebase real-time sync and optimized networking for user-facing control workflows.
- Designed cross-platform UX for booking and payment states, iterating with stakeholders to improve confirmation speed by **40%** across 50K+ records.

Skills

Languages: **Java**, **Kotlin**, Python, C++, JavaScript/TypeScript, Swift (familiar)
Systems: Concurrency, multithreading, memory management, networking, OS fundamentals, low-latency APIs
Applications: Full-stack development, **React Native**, React, REST/WebSockets, UX iteration, code/design review

Projects

AssistControl | *Kotlin, Java, Real-Time Signal Processing*

- Built an assistive control prototype mapping high-frequency input streams to UI actions using **low-latency decoding algorithms**, enabling sub-100ms cursor and selection responses in Android clients.

CPU Scheduling Simulator Live | *C++, React.js, OS Algorithms*

- Implemented preemptive scheduling algorithms with interactive controls and real-time metric visualization, demonstrating first-principles concurrency and resource-management tradeoffs.

FinNexus Live | *TypeScript, WebSockets, Full-Stack UX*

- Shipped a user-facing fintech platform with real-time transaction updates for 50K+ users, architecting reliable, maintainable interfaces under high-volume networking load.